Etienne Gauthier

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Summary

I am a third-year PhD student in the Sierra team at Inria and ENS Ulm, under the supervision of Francis Bach and Michael I. Jordan. My research focuses on decision-making in multi-agent environments, an area that lies at the intersection of machine learning, statistics, and game theory. I expect to graduate in September 2027, but my timeline is flexible.

Education

2024 - 2027 PhD (Computer Science) at **Inria & ENS Ulm**
2023 - 2024 Master's Degree (Mathematics, Vision, Learning) at **ENS Paris-Saclay**
2020 - 2024 Cycle Ingénieur at **École polytechnique**
2018 - 2020 Classes Préparatoires aux Grandes Écoles at **Lycée privé Sainte-Geneviève**

Work Experience

Research Intern, Kyoto University Institute for Advanced Study Mar 2023 – Aug 2023
Kyoto, Japan
- Conducted research under the supervision of Prof. Yasuaki Hiraoka.
- Designed a new kernel for commutative ladders in multidimensional topological data analysis.
- Conducted research in theory of representations of associative algebras.

Annotator, Meta FAIR Aug 2022 – Mar 2023
Paris, France
- Created and validated Lean annotations of mathematical problems from the International Mathematical Olympiad for use in neural theorem proving.
- Supported dataset development for training and evaluating ML models on formal reasoning tasks.

Quantitative Research Intern, Fintica Jun 2022 – Sep 2022
Tel Aviv, Israel
- Developed a Monte Carlo algorithm to identify optimal rolling window sizes, balancing model accuracy and z-score/return similarity.
- Designed a machine learning-based equity investment strategy that outperformed the original approach on key performance indicators.
- Integrated insights from statistical analysis to locate and address suboptimal model performance areas.

Officer Cadet, French Navy Nov 2020 – Apr 2021
***Charles de Gaulle* Aircraft Carrier**
- Completed first-year military training as part of the École polytechnique curriculum.
- Served in an NH90 helicopter unit aboard the *Charles de Gaulle* aircraft carrier.
- Deployed in the Mediterranean Sea, Red Sea, Indian Ocean, and Persian Gulf during the *Clemenceau 21* mission.

Teaching Experience

Teaching Assistant, Université Paris 1 Panthéon-Sorbonne Sep 2022 – Jan 2023

Statistics: Probabilities for 2nd year undergraduate students.

Selected Honors

Google PhD Fellowship in Machine Learning and ML Foundations Awarded November 2025, $140,000 funding

ICML 2025 Oral Presentation and Spotlight Paper Competitive oral session highlighting top papers

Publications

Etienne Gauthier*, Francis Bach, and Michael I. Jordan. "Explaining and Preventing Alignment Collapse in Iterative RLHF". In: *Advances in Neural Information Processing Systems* (2026).

Ben Chugg*, Etienne Gauthier*, Michael I. Jordan, Aaditya Ramdas, and Ian Waudby-Smith. "Post-Hoc Large-Sample Statistical Inference". In: *arXiv preprint arXiv:2603.08002* (2026).

Etienne Gauthier*, Francis Bach, and Michael I. Jordan. "Anytime Detection of Strategic Deviations in Multi-Agent Systems". In: *International Conference on Machine Learning* (2026).

Etienne Gauthier*, Francis Bach, and Michael I. Jordan. "Adaptive Coverage Policies in Conformal Prediction". In: *International Conference on Artificial Intelligence and Statistics* (2026).

Etienne Gauthier*, Francis Bach, and Michael I. Jordan. "Backward Conformal Prediction". In: *Advances in Neural Information Processing Systems* (2025).

Etienne Gauthier*, Francis Bach, and Michael I. Jordan. "E-Values Expand the Scope of Conformal Prediction". In: *arXiv preprint arXiv:2503.13050* (2025).

Etienne Gauthier*, Francis Bach, and Michael I. Jordan. "Statistical Collusion by Collectives on Learning Platforms". In: *International Conference on Machine Learning* (2025). **Accepted for Talk (top 1%)**.

Hideto Asashiba*, Etienne Gauthier*, and Enhao Liu*. "Interval Replacements of Persistence Modules". In: *Discrete & Computational Geometry* (2026).

Talks & Presentations

- Talk: **Anytime Detection of Strategic Deviations in Multi-Agent Systems**, *OCEAN Retreat*, Durham University, UK, Sep 2026.
- Poster: **Anytime Detection of Strategic Deviations in Multi-Agent Systems**, *Crossroads of AI & Society Workshop*, Paris, France, Jul 2026.
- Poster: **Anytime Detection of Strategic Deviations in Multi-Agent Systems**, *International Conference on Machine Learning, Workshop: Statistical Frameworks for Uncertainty in Agentic Systems*, Seoul, South Korea, Jul 2026.
- Poster: **Anytime Detection of Strategic Deviations in Multi-Agent Systems**, *International Conference on Machine Learning*, Seoul, South Korea, Jul 2026.
- Talk: **Anytime Detection of Strategic Deviations in Multi-Agent Systems**, *4th Annual Workshop on Game-Theoretic Statistical Inference*, University of Twente, Netherlands, Jun 2026.
- Poster: **Adaptive Coverage Policies in Conformal Prediction**, *International Conference on Ar-*

tificial Intelligence and Statistics, Tangier, Morocco, May 2026.

- Talk: **Recent Advances in Conformal Prediction with E-Values**, *University of Oxford, Department of Statistics, OxCSML Seminar*, Oxford, UK, Mar 2026.
- Talk: **E-Values Expand the Scope of Conformal Prediction**, *ETH Zürich, Automatic Control Laboratory*, Remote, Feb 2026.
- Talk: **Recent Advances in Conformal Prediction with E-Values**, *Institut de mathématiques de Toulouse, Séminaire de Statistique et Optimisation*, Toulouse, France, Feb 2026.
- Talk: **Adaptive Coverage Policies in Conformal Prediction**, *Joint Meetings of 2025 Taipei International Statistical Symposium and the 13th ICSA International Conference*, Taipei, Taiwan, Dec 2025.
- Talk: **Recent Advances in Conformal Prediction with E-Values**, *Max Planck Institute for Software Systems*, Remote, Dec 2025.
- Poster: **Backward Conformal Prediction**, *Advances in Neural Information Processing Systems*, San Diego, USA, Dec 2025.
- Poster: **Backward Conformal Prediction**, *NeurIPS@Paris*, Paris, France, Dec 2025.
- Talk: **Recent Advances in Conformal Prediction with E-Values**, *Séminaire Parisien de Statistique de l'Institut Henri Poincaré*, Paris, France, Oct 2025.
- Talk: **Recent Advances in Conformal Prediction with E-Values**, *OCEAN Retreat*, Paris, France, Jul 2025.
- Talk & Poster: **Statistical Collusion by Collectives on Learning Platforms**, *International Conference on Machine Learning*, Vancouver, Canada, Jul 2025.
- Talk: **E-Values Expand the Scope of Conformal Prediction**, *3rd Annual Workshop on Game-Theoretic Statistical Inference*, Chennai, India, Jul 2025.
- Talk: **Statistical Collusion by Collectives on Learning Platforms**, *From Matching to Market: A Tale of Mathematics, Economics and Computer Science*, Cargèse, France, Apr 2025.
- Talk: **Statistical Collusion by Collectives on Learning Platforms**, Chaire *Marchés et Apprentissage*, Paris, France, Mar 2025.

Academic Service

Reviewer (Journals): Journal of the Royal Statistical Society Series B, Biometrika, Transactions on Machine Learning Research

Reviewer (Conferences): ICML 2026 (Gold Reviewer), NeurIPS 2026, AISTATS 2026